Fall 1994

Problem 1 (Fa94). For which values of the real number a does the series

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \sin \frac{1}{n} \right)^{a}$$

converge?

Problem 2 (Fa94, Sp24). Prove that the matrix

$$\begin{pmatrix} 1 & 1.00001 & 1 \\ 1.00001 & 1 & 1.00001 \\ 1 & 1.00001 & 1 \end{pmatrix}$$

has one positive and one negative eigenvalue.

Problem 3 (Fa94). Evaluate the integrals $\displaystyle\int_{-\pi}^{\pi} \frac{\sin n\theta}{\sin \theta}\,\mathrm{d}\theta, \quad n = 1, 2, \ldots$

Problem 4 (Fa94). Suppose the group G has a nontrivial subgroup H which is contained in every nontrivial subgroup of G. Prove that H is contained in the center of G.

Problem 5 (Fa94).

1. Find a basis for the space of real solutions of the differential equation

$$\sum_{n=0}^{7} \frac{\mathrm{d}^n x}{\mathrm{d}t^n} = 0$$

2. Find a basis for the subspace of real solutions that satisfy $\lim_{t \to +\infty} x(t) = 0$

Problem 6 (Fa94). Let $A = \left(a_{ij}\right)_{i,j=1}^{n}$ be a real $n \times n$ matrix such that $a_{ii} \geqslant 1$ for all i, and

$$\sum_{i \neq j} a_{ij}^2 < 1$$

Prove that A is invertible.

Problem 7 (Fa94). Let f be a continuously differentiable function from $\mathbb{R}^2$ into $\mathbb{R}$. Prove that there is a continuous one-to-one function g from [0, 1] into $\mathbb{R}^2$ such that the composite function $f \circ g$ is constant.

Problem 8 (Fa94). Let $\mathbb{Q}$ be the field of rational numbers. For θ a real number, let $\mathbb{F}_\theta = \mathbb{Q}(\sin\theta)$ and $\mathbb{E}_\theta = \mathbb{Q}(\sin\theta/3)$. Show that $\mathbb{E}_\theta$ is an extension field of $\mathbb{F}_\theta$, and determine all possibilities for $\dim_{\mathbb{F}_\theta} \mathbb{E}_\theta$.

Problem 9 (Fa94). Evaluate $\displaystyle\int_0^\infty \frac{\log^2 x}{x^2+1}\,\mathrm{d}x$.

Problem 10 (Fa94). Let the function $f : \mathbb{R}^n \to \mathbb{R}^n$ satisfy the following two conditions:

- $f(K)$ is compact whenever K is a compact subset of $\mathbb{R}^n$.
- If $\{K_n\}$ is a decreasing sequence of compact subsets of $\mathbb{R}^n$, then

$$f\left(\bigcap_1^\infty K_n\right) = \bigcap_1^\infty f(K_n)$$

Prove that f is continuous.
Note: See also **????**.

Problem 11 (Sp77, Sp93, Fa94). Find a list of real matrices, as long as possible, such that

- the characteristic polynomial of each matrix is $(x-1)^5(x+1)$,
- the minimal polynomial of each matrix is $(x-1)^2(x+1)$,
- no two matrices in the list are similar to each other.

Problem 12 (Fa94, Sp17). Suppose the coefficients of the power series $\displaystyle\sum_{n=0}^\infty a_n z^n$ are given by the recurrence relation

$$a_0 = 1 \quad a_1 = -1 \quad 3a_n + 4a_{n-1} - a_{n-2} = 0 \qquad n = 2, 3, \ldots$$

Find the radius of convergence of the series and the function to which it converges in its disc of convergence.

Problem 13 (Fa94, Fa21). Let p be an odd prime and $\mathbb{F}_p$ the field of p elements. How many elements of $\mathbb{F}_p$ have square roots in $\mathbb{F}_p$? How many have cubic roots in $\mathbb{F}_p$?

Problem 14 (Fa94, Sp98, Sp15). Find the maximum area of all triangles that can be inscribed in an ellipse with semiaxes a and b, and describe the triangles that have maximum area.
Note: See also **??**.

Problem 15 (Fa94). Let A be a diagonal matrix in $M_{7\times 7}$ that has $+1$ in four diagonal positions and -1 in three diagonal positions. Define the linear transformation T on $M_{7\times 7}$ by $T(X) = AX - XA$. What is the dimension of the image of T?

Problem 16 (Fa94). Let $\mathbb{D}$ denote the open unit disc in $\mathbb{R}^2$. Let u be an *eigenfunction* for the Laplacian in $\mathbb{D}$; that is, a non identically zero, real valued function of class C^2 defined in $\overline{\mathbb{D}}$, such that

$$\begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = \lambda u & \text{in } \mathbb{D} \\ u = 0 & \text{in } \partial\overline{\mathbb{D}} \end{cases}$$

where λ is a constant. Prove that

$$\iint_{\mathbb{D}} \|\nabla u\|^2 \,\mathrm{d}x\,\mathrm{d}y + \lambda \iint_{\mathbb{D}} u^2 \,\mathrm{d}x\,\mathrm{d}y = 0 \qquad (*)$$

and hence that $\lambda < 0$.

Problem 17 (Fa94, Fa13, Sp24). Let R be a ring with identity, and let u be an element of R with a right inverse. Prove that the following conditions on u are equivalent:

1. u has more than one right inverse;
2. u is a zero divisor;
3. u is not a unit.

Problem 18 (Fa94). Let the function f be analytic in the complex plane, real on the real axis, 0 at the origin, and not identically 0. Prove that if f maps the imaginary axis into a straight line, then that straight line must be either the real axis or the imaginary axis.